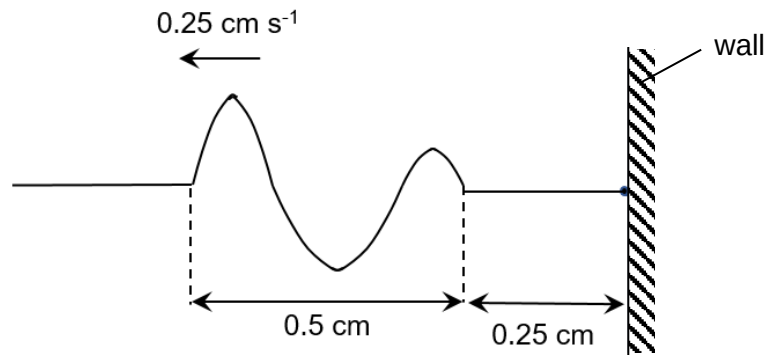


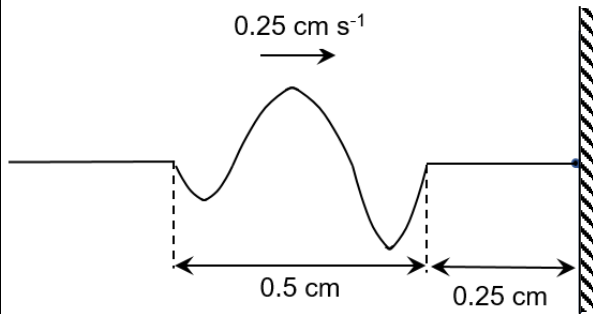
- 16** A rope is rigidly fixed to a wall. A wave pulse of length 0.5 cm moves at a constant speed of 0.25 cm s^{-1} along the rope towards the wall and is reflected back.

The figure below shows the waveform at time $t = 4\text{ s}$.

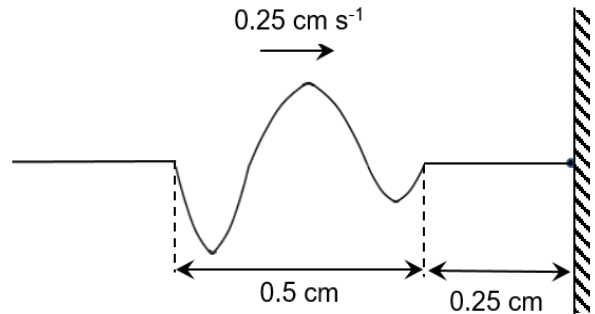


Which of the following correctly shows the waveform at $t = 0\text{ s}$?

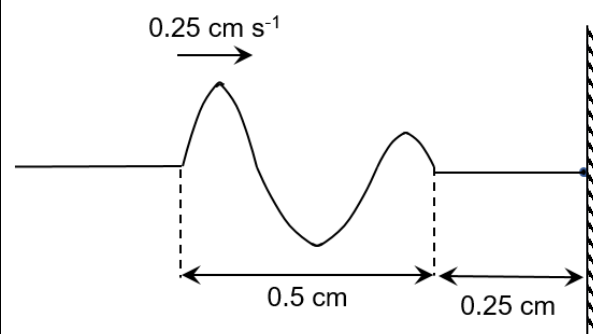
A



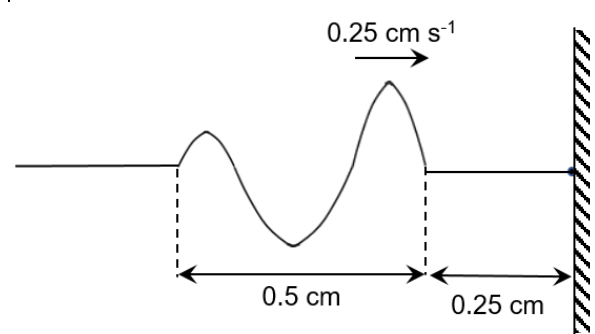
B



C



D



Answer: A

Checking position of the wave pulse:

$$0.5\text{ cm} / 0.25\text{ cm s}^{-1} = 2\text{ s}$$

In 2 s, the wave pulse moves a distance equal to one wavelength.

From options A, B, C, D, which shows the wave pulse at $t = 0\text{ s}$, it means that at $t = 2\text{ s}$, half of the wave pulse would have been reflected, and at $t = 4\text{ s}$, it would have been at the position shown.

Determining the shape of the incident waveform:

Since the wave pulse is **reflected at a fixed end**, the reflected pulse will have a **180° change in phase**. Hence answer is A.

